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Are there time and cost savings by using telemanagement for patients on intensified insulin therapy? A randomised, controlled trial

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Abstract

Background: Patients with insulin dependent diabetes require frequent advice if their metabolic control is not optimal. This study focuses on the fiscal and administrative aspects of telemanagement, which was used to establish a supervised autonomy of patients on intensified insulin therapy. *Methods*: A prospective, randomised trial with 43 patients on intensified insulin therapy was conducted. Travelling distance to the diabetes centre was 50 min one way; all patients had undergone a diabetes education course with lessons in dose adaptation. Patients were randomly assigned to telecare ($n = 27$) or conventional care ($n = 16$). They used BG-meters with a storage capacity of 120 values (Precision QID™ Abbott/Medisense) and transmitted their data over a combined modem/interface via telephone line to the diabetes centre. Data were displayed and stored by a customised software (Precision Link Plus™, Abbott/Medisense). Advice for proper dose adjustment was given by telephone. *Results*: Average time needed for instruction in the telemedical system was 15 min. Data were transmitted every 1–3 weeks and a teleconsultation was performed by phone every 2–4 weeks, depending on the extent of specific problems. On average, personal visits in the control group were performed once a month. Physician's time expenditure for telemanagement, compared to conventional advice was moderately higher (50 vs. 42 min per month). A substantial amount of time on the patients side could be saved through replacing personal communications by telephone contacts and data transmission reduction (96 vs. 163 min/month including data transmission time). Setting up an optimal telemanagement scenario, a cost analysis was carried out yielding savings of ≈ 650 EURO per year per patient. HbA_{1c} dropped significantly from 8.2 to 7.0% after 8 months of observation, but there was no significant difference between the intervention and control groups. Major technical problems with the telematic system did not occur during the study. *Conclusions*: Telemanagement of insulin-requiring diabetic patients is a cost and time saving procedure for the patients and results in metabolic control comparable to conventional outpatient management. © 2002 Elsevier Science Ireland Ltd. All rights reserved.

Keywords: Telemedicine; Diabetes; Telecare; Insulin therapy; Glucose monitoring

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1. Introduction

Modern insulin therapy of patients with type-1 and -2 diabetes mellitus is guided by the results of recent studies concluding that near normoglycemic metabolic control reduces or retards the incidence of late complications of the disease [1]. To achieve optimal metabolic control, the joint efforts of patients, physicians and the care team as a whole are necessary. Best results have been achieved by performing intensified insulin therapy. The principle of this form of insulin therapy basically consists of separate substitution of basal long acting and prandial short acting insulin, using predefined target ranges for blood glucose values and correction limits for adapting the dose of the short acting insulin with every injection. A training program for dose adaptation included in a structured diabetes education program is considered mandatory for patients starting intensified insulin therapy. Because of the near normoglycemic blood glucose targets, hypoglycaemias tend to be more frequent in these patients, partly with subsequent hyperglycemias (Somogyi-effect) and metabolic control may even become unstable, preferentially in long standing disease.

On the other hand, novel insulin preparations must be considered progress towards a better adaptation to the real insulin requirements at night and before meals. Patients are enabled to optimise their insulin therapy by permanently fine-tuning insulin dosage according to varying life conditions, such as physical activity, concurrent illness, shift-working, etc. Although the specific training programs for intensified insulin therapy are designed to give patients autonomy and ability for self-management of the disease, additional supervision by a competent care team is frequently necessary, particularly if the problems mentioned above are present and/or therapy goals are not achieved. This additional supervision and advice is mostly needed for a limited period of time only, until good metabolic control is achieved or re-achieved under daily living conditions and the patient feels confident enough to take over responsibility for the therapy. Consultations with the diabetes specialist can then be tapered to regular 3-monthly or twice yearly routine visits.

A variety of computer applications have been developed for documentation, processing and display of diabetes-data [2]. Technological progress in recent years has made it possible to transfer self-monitored blood glucose values, stored together with date and time in memory meters, to a remote diabetes specialist using simple, modern telecommunication techniques and remote data transfer. Advice can be given to patients without the need for direct personal contact, potentially saving time and expenditure. Repetitive application of these methods in chronic disease is called telecare. Before these techniques can be established as routine methods, however, including reimbursement by the care providers, an evaluation is important regarding medical outcome, cost-efficiency and patient satisfaction/quality of life.

In our study, patients with diabetes mellitus on intensified insulin therapy were divided into two groups by randomisation, one using telecare and the other conventional outpatient care. Each group was under relatively tight medical supervision and therapeutic goals and efforts did not differ between groups in order to achieve optimal metabolic control in all of the patients.

2. Patients and methods

A total of 48 patients on intensified insulin therapy were randomly assigned to two groups, one using telecare and one on conventional care (controls). Criteria for intensified therapy were a minimum of four insulin injections daily, separation of basal and meal-related insulin, a predefined target range for preprandial blood glucose, having undergone a structured diabetes education program, self-control of BG-values before every insulin injection and estimation or calculation of the insulin dose by carbohydrates to be ingested and actual BG-value (dose correction). Characteristics of patients in the two groups are listed in Table 1. Patients on telecare received an additional lesson in handling the modem for data transfer and were handed a technical bulletin with FAQ's and possible error solutions. They received a modem preprogrammed with their ID and the telephone number of the diabetes-centre

Table 1
Patient characteristics/profile

Patients	Telecare	Controls
Number of patients that finished the study (drop outs)	27 (3)	16 (2)
Age (years)	30.5 ± 11	30 ± 8.6
Duration of diabetes (years)	10.9	8.1
Mean travel distance to diabetes centre (min)	50	47

for automated dialling. Patients determined their BG-values by using memory meters, storing these values together with their respective date and time. The meters could be connected to the pre-programmed modem, which automatically uploads the data via the analog telephone system to the diabetes centre. There, the PC with a customised software (Precision Link Plus™, MediSense/Abbott, Wiesbaden, Germany) is permanently in a ready-to-receive status, recognising the data by the patient's ID. New incoming data are marked on the screen to facilitate recognition. The software requires at least a 486 processor, 8 MB of random access memory, 6 MB of hard disk space and can be operated under Windows 95/98™. The telemedical system is shown in Fig. 1, a screenshot of the Precision Link Plus™ software in Fig. 2.

All patients underwent a physical examination, medical history and laboratory control (HbA_{1c}) at the beginning of the study. For safety considerations, regular personal visits—including laboratory controls and metabolic assessment, e.g. frequency of hypoglycaemias—were scheduled every 2 months. Patients were in the study for a minimum of 4 and a maximum of 8 months. In those patients achieving their personal therapy goal (metabolic stability, improvement of HbA_{1c}, reduction of hypoglycaemia) within 4 months, the study was terminated after 4 months. The remaining continued telecare until the end of the 8-month period. At the final visit, another laboratory control and metabolic assessment was performed. Additional consultations, i.e. teleconsultations for the telecare group and personal visits for the controls, were scheduled on demand. The study protocol is shown in Fig. 3. For urgent consultations, patients were able to contact the centre via a 24-h voice recorder system and a consultation with the physician was established on the following day. Telecare patients were asked to transmit their BG-values before each personal visit or telephone consultation, or, because of the limited storage capacity of the BG-meters, at least every 2 weeks.

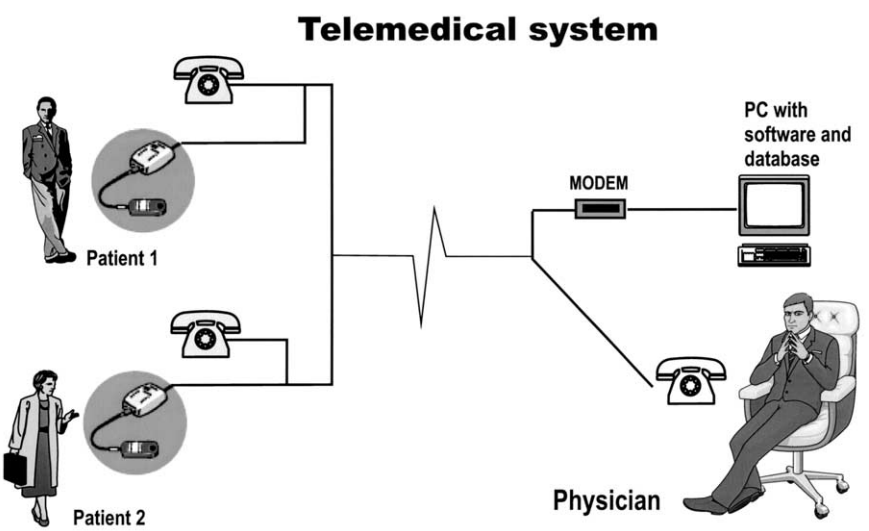

Fig. 1. The telemedical system.

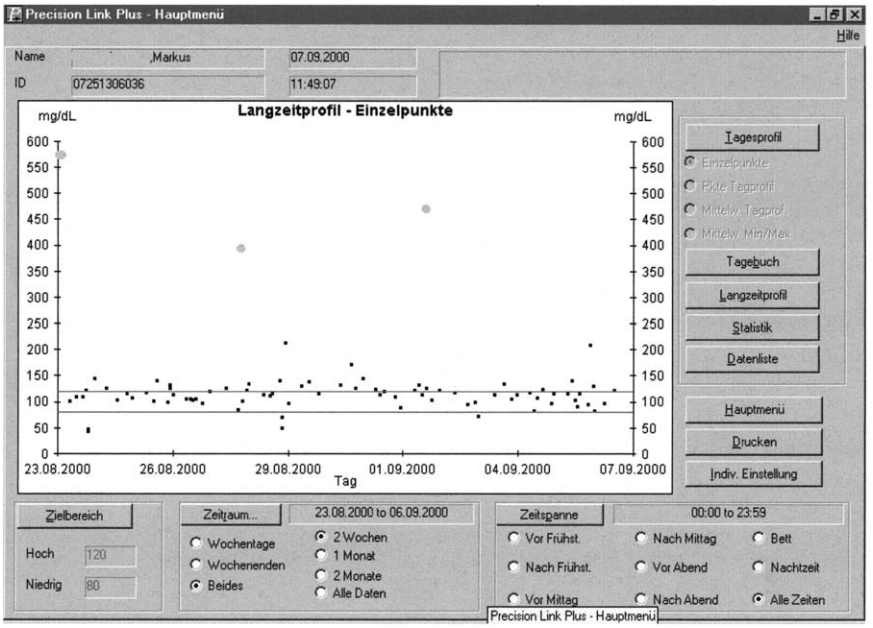

Fig. 2. Screenshot of the 'Precision link plus' software used in the office of the care team. This window displays the blood glucose values of a patient for 14 days as single points.

The time required for patient management was documented by the physicians with every personal or telephonic patient contact. Time required for the patients, including travel and data transmission times, was documented by questionnaire. The questionnaire also contained questions regarding patient satisfaction and proposals for improvements of the telemedical system.

2.1. Statistical methods

Randomisation was carried out by lots with a chance of 2:1 in favour of telecare. Experience has shown difficulties in recruiting patients for the control group if their chance is only 50% to get into the treatment group. The loss in statistical power with different sample sizes was therefore tolerated. Two sided Student's t-test was used for statistical analysis of HbA_{1c}-values, which were found to be almost normally distributed. The needed sample size was calculated assuming a delta in HbA_{1c} of 1%, a S.D. of 1.5%, a type 1 error of 0.05 (95% confidence) and a type 2 error of 0.1.

3. Results

3.1. Patient profile

A total of 48 patients were included in the study, 30 on telecare and 18 in the control group. Average age was 30.5 ± 11 years for telecare patients and 30.0 ± 8.6 years for controls, average duration of the diabetes 10.9 and 8.1 years, respectively. Travel time to the diabetes center from the patients home varied greatly from 5 min to 2 h one-way and averaged 50 min for the telecare group and 47 min for the controls. HbA_{1c} at the beginning of the study was $8.3 \pm 2.3\%$ in the telecare group and $8.0 \pm 2.1\%$ in the control group (n.s.). For additional metabolic parameters see Table 1. These data indicate fairly good matching of the patients between the two groups. The differences in the frequency of hypoglycaemia are random due to the wide dispersion of these values and the relatively small sample size (see remarks below).

Further characteristics, such as metabolic stability, instability or recent diabetes diagnosis etc.,

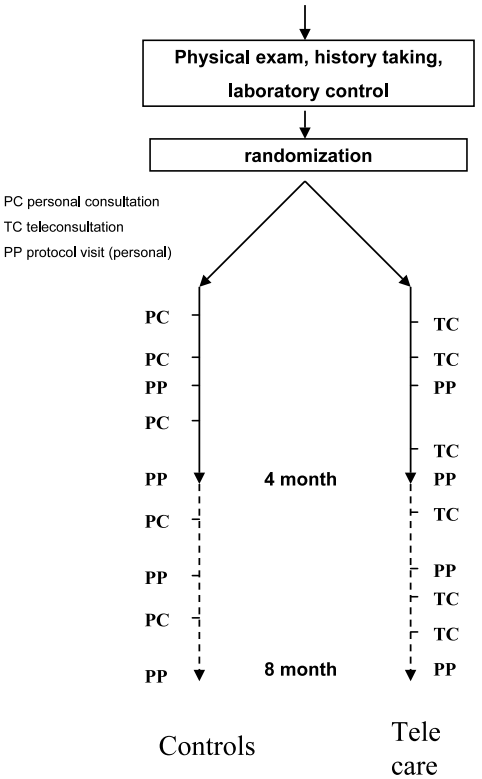

Fig. 3. Study protocol: the personal visits (PP) are scheduled at fixed intervals every 2 months. All other visits (PC) or teleconsultations (TC) are variable.

are listed in Table 2. Due to the frequency of multiple problems in single patients, patients could not easily be classified in only one of the categories. Therefore, further results of patients in these subgroups are not given in detail here. There was a tendency though, towards a higher glycosylated haemoglobin level, higher frequency of hypoglycaemia, more frequent physician contacts during the study and more time spent on consultations in patients classified as 'metabolically unstable' compared with the other groups. Five patients did not complete at least 4 months in the study and were regarded as drop outs. One patient in the telecare group had problems with technical handling the system, two young patients had compliance problems not related to data transfer. Two persons in the control group were non-compliant with regular appointments.

Table 2
Subgroups according to specific problems

Group	Telecare	Controls
New onset diabetes	7	3
Stable metabolic control	7	7
Unstable metabolic control	9	3
Shift worker	4	1
Not to assign	1	2

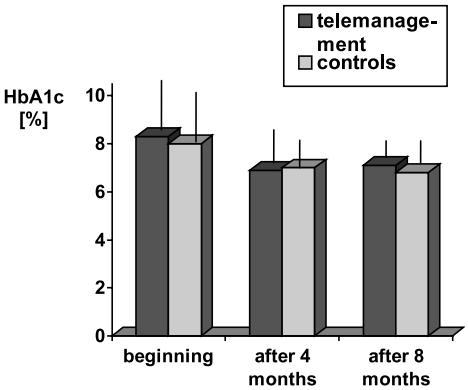

Fig. 4. HbA_{1c}-values ($\pm$S.D.) of patients with telecare ($n = 27$ after 4 months and $n = 11$ after 8 months) and conventional care ($n = 16$ after 4 months and $n = 10$ after 8 months) in the beginning. There was no significant difference between the groups, indicating that medical outcome in terms of metabolic control is comparable.

3.2. Metabolic outcome

Glycosylated haemoglobin levels improved in both groups to $6.9 \pm 1.3\%$ after 4 months (27 patients) and $7.1 \pm 0.7\%$ after 8 months (11 patients) in the telecare group and $7.0 \pm 1.0\%$ after 4 months (16 patients) and $6.8 \pm 1.1\%$ after 8 months (ten patients) in the control group. Differences between the groups were not statistically significant (Fig. 4). For applying statistical tests on the comparison of the other metabolic parameters, hypoglycaemia and insulin dose, sample size was too small. However, both the number of hypoglycaemias and average insulin requirement tended to converge during the study with differences between control and intervention groups being smaller in the end (Fig. 5). This result probably reflects the general better metabolic control in both groups at the end of the study.

3.3. Time requirements and fiscal analysis

Telecare patients conducted an average of 1.75 data transfers per month, each taking 4.2 min including all preparations necessary (range 1–15, questionnaire results). A total of 521 data transfers were carried out. Problems with data transfers arose in $< 5\%$ of the data transfers and could mostly be solved by simply repeating the transfer. Problems that needed more than 5 min to solve arose in $< 1\%$ of all data transfers.

Patients of the telecare group spent an average total of 554 min (range: 220–1056) for their consultations after 4 months and of 974 min (range: 399–1762) after 8 months. In the control group, these times were 656 min (range: 255–1745) after 4 months and 959 min (range: 374–3105) after 8 months. The times include travel, waiting, consultation, data transmission times and times for completing the study protocol in all patients. Consultation times were taken from the patients chart records and all other times were recorded by questionnaire. For better comparison, average times per month are depicted in Table 3.

In order to make fiscal calculations under mere telemedical conditions, the mandatory 2-monthly personal visits required by the study protocol out of safety considerations were replaced by average telephone consultation times for accounting purposes (Table 3, line 2). Patients in the telecare group needed less time for their consultations than controls (83 vs. 163 min per month for the 4-month period and 49 vs. 119 min per month for the 8-month period), although they had more contacts with their physician (average 7.3 (range 3–13) vs. 5.3 (range 3–10) contacts for the 4-month period and average 13.5 (range 7–20) vs. 8.6 (range 5–17) contacts for the 8-month period) (Table 3). These differences mostly reflect differences in travel times, which were greater due to more frequent travel to the diabetes center in the control group and, to a lesser part, shorter consultation times via telephone and data transmission in the telecare group.

Time required for telecare patients on the physician's side was significantly higher than time needed for the care of control patients (50 vs. 43 min per month for the 4-month period and 43 vs. 34 min per month for the 8-month period), reflecting the higher number of patient contacts rather than time spent on handling the telecare system.

Data collected in the questionnaires included travel distance and time to the diabetes center. A cost analysis was carried out estimating average travel costs and average telephone costs by distance for all patients. An estimation of the total costs of telecare in an optimised scenario, including a comparison to conventional care, is discussed below.

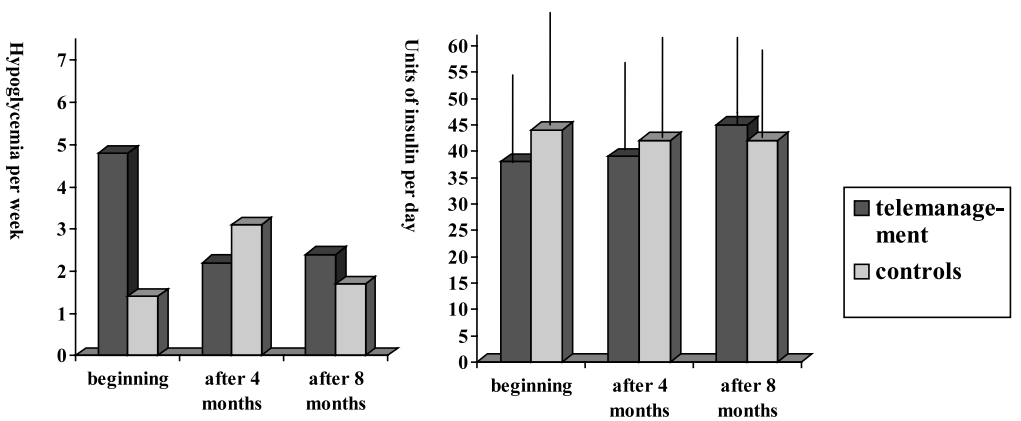

Fig. 5. Additional metabolic parameters in the beginning, after 4 months and after 8 months. (A) Frequency of hypoglycaemic events per week in patients with telemanagement and in controls. (B) Average units of insulin per day ($\pm$S.D.) for both groups.

Table 3
Times required for the health care professional (1) and the patients (2+3) for telecare and conventional care in minutes per month as mean values

Time required for (min per month)		Within 4 months		Within 8 months	
		Telecare ($n = 27$)	Controls ($n = 16$)	Telecare ($n = 11$)	Controls ($n = 10$)
1	Physician	50	43	43	34
2	Patients excluding time for study protocol	83	163	49	119
3	Patients including time for study protocol	138	164	121	119

Patient's time requirements comprise the data revision process and advice by telephone, personal consultations for instruction in the use of the modem and data transmission (2) and additionally (3) time needed to complete the study protocol (every 2 months) and time needed for data transmissions.

3.4. Patient satisfaction

The patient questionnaire contained questions about patient satisfaction with the telemedical system and telecare. Out of 27 telecare patients, 23 (85%) felt that telecare was better than conventional care, reasons being better surveillance of BG-data by the physician, faster intervention in the case of problems and no waiting and travel times for the consultations. An additional reason for preferring telecare was a feeling of security on the patients side through the availability of all BG-data for the physician and the possibility of consulting the physician within minimal time without the need to travel to the diabetes center, a point especially important, e.g. in concurrent illness (personal communications of the patients). Some 15% reported difficulties in handling the telecare system, 85% found it easy to use. Asked whether they would enter additional data such as insulin doses, carbohydrate units or hypoglycaemia if required by the physician, patients answered as follows (multiple answers permitted):

26% in any case
30% if not time consuming
21% if the daily log book can be abolished
9% only in periods with poor metabolic control
14% not at all

Patients were asked whether they conducted a conventional diabetes diary (with documentation of BG-values, ingested carbohydrate units and insulin doses) on paper (multiple answers permitted):

no, I found it bothersome	14%
yes, although I found it bothersome	13%
not regularly, only when I had problems	9%
yes, but my doctor did not look at it	9%
yes, because it gave me security	59%
yes, my doctor looked at it regularly	63%

4. Discussion

Novel care methods for diabetic patients require an evaluation [3]. If new elements due to the booming market of telecommunication are added to the care process, various problems may arise. First, the new method may be more costly than conventional medical care, specialised technical know-how is usually required and the technical equipment needed is not normally stored in average households. Furthermore, the trustful and confidential relationship between patient and doctor may be harmed by using distance-communication methods and maintaining only a minimum (or no) personal contact during the problem solving process. This pilot study was undertaken to

address these questions, testing the practicability of telecare in routine patient management, including fiscal considerations and to potentially demonstrate technological and logistic improvements of the care process with telemedical devices.

Telemedical management improved glycosylated haemoglobin of the patients in this study, but in this respect was not significantly better than conventional care in the control group. This result is most probably a consequence of the relatively intensive specialist care provided for patients in both groups, either by telecare, or by personal contact. The professional skills in supervision of patients with IDDM are expressed by an amelioration of HbA_{1c} and/or a reduction of the frequency of hypoglycaemia by redistribution and in case of a possible increase of insulin dose. The results of this study regarding hypoglycaemias and insulin requirements display this tendency, however, sample size was too small to calculate a statistical significance.

Other investigators have found, as far as control groups were included in the study design, varying differences between telecare patients and controls regarding metabolic outcome, e.g. lower HbA_{1c} in the telecare group. In our opinion these differences are probably more related to the intensity of the specialists involvement in the patients therapy than to telecare itself. In a simplified telemedicine setting with telephone consultations, Thompson et al. found an improvement in the HbA_{1c} level of their telephone care patients from 9.6% at baseline to 7.8% after 6 months, whereas control patients did not improve significantly. There was no intensification of standard care, however, to improve metabolic status of patients in the control group, compared with intensification of care in the telemedicine group [4]. Other studies describe varying degrees of metabolic improvement in their telecare patients, without inclusion of control groups. Liesenfeld et al. observed a reduction in the frequency of hypoglycaemias in children and adolescents [5]. A reduction of clinical visits from six to three per year in telecare patients was reported by Meneghini et al. [6]. However, the technology used is based on a day-by-day advisory system with a semi-automatic advice. However, the time expenditure per month and patient on the side of the health care professionals in the study was comparable to our calculations. There was a greater decline of glycosylated haemoglobin in the study of Salzsieder et al. [7], but there is also no comparison to a control group. In contrast to the implicated findings of these authors, our study suggests that the improvement of metabolic status in telecare patients is rather due to the intensification of specialist consultations than to telecare itself. Hence, it is strongly dependent on the study design whether an effect of therapeutic intensification or of telemedicine application itself is shown.

We believe that net improvements of some metabolic parameters will always be observed once a specialist is involved in the care process and in order to extract the possible benefits attributed to telecare, a control group should be added, as was the case in our study. In the current stage, we believe that telecare in diabetes is best suited as a problem solving tool rather than for continuous care monitoring, if metabolic parameters are in the target range. Consequently, a crossover design in an evaluation study would not be ideal to prove its efficacy.

In order to arrive at conclusive results all contacts and visits were recorded, including the time needed (Table 3). To transform these data into costs we set up a scenario transforming all documented times and distances into costs under the preconditions shown in Table 4. Since time spent on physicians attendance did not differ much between groups and physician costs are difficult to compare internationally, differences in physicians' cost have been disregarded. All times were extrapolated to 1 year of continuous care.

It could be demonstrated that the overwhelming effect of telemedicine in diabetic patients with intensified insulin therapy is due to time and cost savings on the patients side, namely travel time and workplace absence at the time of personal medical consultation, this effect being more pronounced for patients living in remote areas. Costs for the technical device and telephone and data transfer are highly compensated by these savings. Certain subgroups have more benefit from telecare than others. Patients with new onset type-1-diabetes entering the remission phase are

confronted with changing insulin needs and are inexperienced with dose adaptations. Diabetics with frequently changing office hours or shift-workers require particular attention because meal and injection times must be adapted to their personal needs. Switching from one insulin preparation to another, such as from regular insulin to lispro or aspart insulin is a typical situation in which telecare is helpful and hospitalisation can be avoided, even if metabolic control is not optimal.

Which significance is expected from telemedicine for diabetic patients in the future? Even if diabetes education is improved and any patient will have access to it, there will always be periods when patients with IDDM are running out of metabolic control and close medical supervision is needed. Furthermore, it can be expected that novel insulin preparations, new ways of administration and new methods for self-control (including continuous glucose sensing) are established and require know how and assistance. Documentation with paper and pen will continuously be replaced by electronic documentation using improved glucose meters and probably, in rising numbers of households, the personal or palmtop computers. The desire for more mobility for our patients in the future might even lead to mobile telephones uploading diabetes-related data to the diabetes centre.

Our survey of the patients participating in the study has shown that one-third found it bothersome to conduct a daily log book with pen and paper. Novel and user-friendly devices are expected to appear on the market to make this task redundant. However, the options to enter, store and visualise blood glucose values are momentarily more advanced than they are for insulin doses and diabetes-related events. The available software designed to accompany possible consultations has improved, but has not yet achieved the degree of desired flexibility for fast and action oriented consultations.

On the side of health care professionals, telecare has been 20% more time consuming than conventional care. Due to more convenient access to the physician, patients tended to call more often. The limiting factors on this side of the line in the future are: the complexity of the technology and the reimbursement of the health care providers.

Which form of telecare will prevail in the future? Several methods have been described in literature: day-by-day and visit-by-visit [8]. Regarding the choice of the method, we believe that it is essential to obtain an efficient and flexible procedure both for the patient and for the health care professional. Mode, time and time period between teleconsultations should be based on mutual arrangement. However, an ancillary condition must be the maximal available time that will be reimbursed by the health care provider. Rigid time schedules have not been established in our pilot study. Particularly user-friendly consultation time in the after office hours were highly asked. Whether semi-automatic devices with day-by-day

Table 4
Scenario for a cost-comparison analysis

Costs per year (Euro)	Leasing modem	Phone consultations	Data transfer	Travel costs	Costs for not working	Subtotal
Telemanagement	185.00	22.00	9.00	45.00	128.00	389.00
Conventional				270.00	767.00	1037.00
Difference attributable to telecare	185.00	22.00	9.00	−225.00	−639.00	−648.00

The boundary conditions and assumptions are based on 1999 costs for the German telecom and public transportation system. Presumptions: Eu: 0.23 for a data transfer of 2 min duration; Eu 1.60 for a telephone call of 15 min duration; costs for not working 2 h 30 min × Eu 25 = Eu 62.5 per consultation; travel expenses for 50 min = Eu 23.0; costs for modem leasing = Eu 16.0 per month. Twenty data transfers per year, two personal visits of telecare patients per year; ten telephone consultations per year. Conventional care: one personal visit per month on average.

knowledge-based or rule-based advisory systems [8–11] can reduce time expenditure in the health care team and further improve metabolic outcome, is currently under investigation. Because of the fast spreading even among patients with diabetes, internet-based applications seem to be an attractive method for communication [12].

5. Conclusion

Telemanagement of insulin-requiring diabetic patients with a preprogrammed modem on the patients side and a PC with a customised software in the health professionals office is a cost and time saving procedure for the patients and results in metabolic control comparable to conventional outpatient management.

Acknowledgements

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References

[1] The Diabetes Control and Complication Trial Research Group, The effect of intensive treatment of diabetes on the development and progression of long term complications in insulin-dependent diabetes mellitus, New Engl. J. Med. 329 (1993) 977–986.

[2] E.D. Lehmann, T. Deutsch, Applications of computers in diabetes care—a review, Med. Inform. 20 (1995) 281–302 and pp. 303–329.

[3] H. Burchert, J.-U. Müller, Zur Ökonomie telemedizinischer Netzwerke, in Jäckel (Hrsg.), Telemedizinführer Deutschland, Bad Nauheim, Ausgabe 2000.

[4] D.M. Thomson, S.F. Kozak, S. Sheps, Insulin adjustment by a diabetes nurse educator improves glucose control in insulin-requiring diabetes patients: a randomised trial, CMAJ 161 (8) (1999) 959–962.

[5] B. Liesenfeld, R. Renner, M. Neese, Telemedical care reduces hypoglycemias and improves glycemic control in children and adolescents with Type 1 diabetes. Diabetes Technol. Ther. 2(4) (2002) (Winter) 561–567.

[6] L.F. Meneghini, R.B. Goldberg, A.M. Albisser, D.H. Mintz, An electronic case manager for diabetes control, Diabetes Care 21 (4) (1997) 591–596.

[7] E. Salzsieder, H. Bolz, J.C. Ramirez, A. Rutscher, U. Fischer, Ch. König, Telematics in diabetes care: application of an advisory program based on routinely monitored self control data, Diabetologia 42 (Suppl. 1) (1999) A54.

[8] T. Deutsch, E.D. Lehmann, E.R. Carson, A.V. Roudsari, K.D. Hopkins, P.H. Sönksen, Time series analysis and control of blood glucose levels in diabetic patients, Comp. Methods Prog. Biomed. 41 (1994) 167–182.

[9] E. Biermann, W. Heinlein, E. Standl, Computer assisted analysis of self-monitoring blood glucose and insulin values from IDDM patients with intensified insulin therapy, Proc. Symp. Comp. Diabetes Graz 31 (1996) Abstract.

[10] St. Montani, R. Bellazi, C. Larizza, A. Riva, C. d'Annuncio, St. Fiocci, R. Lorini, M. Stefanelli, Protocol-based reasoning in diabetic patient management, Inter. J. Med. Informat. 53 (1999) 61–77.

[11] M.E. Hernando, E.J. Gomez, F. del Pozo, R. Corcoy, DIABNET: a qualitative model-based advisory system for therapy planning in gestational diabetes, Med. Inform. (Lond.) 21 (4) (1996) 359–374.

[12] R. Bellazi, S. Montani, A. Riva, M. Stefanelli, Web-based telemedicine systems for home-care: technical experiences, Comp. Methods Progr. Biomed. 64 (3) (2001) 175–187.